

Smart Home Automation System

Complete ESP32 Embedded C++ Firmware Implementation (MQTT & Sensor Integration)

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// SmartHomeAutomation.ino
```

```
#include <WiFi.h>
#include <PubSubClient.h>
#include "DHT.h"

// =====
// Hardware Configuration & Pin Mappings
// =====
#define DHTPIN 15
#define DHTTYPE DHT11
#define LDRPIN 34
#define PIRPIN 13

#define RELAY1_PIN 18
#define RELAY2_PIN 19
#define RELAY3_PIN 21
#define RELAY4_PIN 22

// =====
// Network & MQTT Broker Credentials
// =====
const char* ssid = "YOUR_WIFI_SSID";
const char* password = "YOUR_WIFI_PASSWORD";
const char* mqtt_server = "broker.hivemq.com";

WiFiClient espClient;
PubSubClient client(espClient);
DHT dht(DHTPIN, DHTTYPE);

unsigned long lastMsg = 0;

// =====
// Wi-Fi Connection Setup
// =====
void setup_wifi() {
  delay(10);
  Serial.print("Connecting to ");
  Serial.println(ssid);
  WiFi.begin(ssid, password);

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
```

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    Serial.print(".");
}

Serial.println("
WiFi connected. IP address: ");
Serial.println(WiFi.localIP());
}

// =====
// MQTT Inbound Command Callback
// =====
void callback(char* topic, byte* message, unsigned int length) {
    String messageTemp;
    for (int i = 0; i < length; i++) {
        messageTemp += (char)message[i];
    }

    Serial.print("Message arrived on topic: ");
    Serial.print(topic);
    Serial.print(". Message: ");
    Serial.println(messageTemp);

    if (String(topic) == "home/relay1") {
        digitalWrite(RELAY1_PIN, messageTemp == "ON" ? LOW : HIGH);
    } else if (String(topic) == "home/relay2") {
        digitalWrite(RELAY2_PIN, messageTemp == "ON" ? LOW : HIGH);
    } else if (String(topic) == "home/relay3") {
        digitalWrite(RELAY3_PIN, messageTemp == "ON" ? LOW : HIGH);
    } else if (String(topic) == "home/relay4") {
        digitalWrite(RELAY4_PIN, messageTemp == "ON" ? LOW : HIGH);
    }
}

// =====
// MQTT Broker Reconnection Loop
// =====
void reconnect() {
    while (!client.connected()) {
        Serial.print("Attempting MQTT connection...");
        if (client.connect("ESP32SmartHomeClient")) {
            Serial.println("connected");
            client.subscribe("home/relay1");
            client.subscribe("home/relay2");
            client.subscribe("home/relay3");
            client.subscribe("home/relay4");
        } else {
            Serial.print("failed, rc=");
            Serial.print(client.state());
            Serial.println(" try again in 5 seconds");
            delay(5000);
        }
    }
}

// =====
// System Setup Initialization
// =====
void setup() {
    Serial.begin(115200);

```

```

// Initialize Relay Outputs
pinMode(RELAY1_PIN, OUTPUT);
pinMode(RELAY2_PIN, OUTPUT);
pinMode(RELAY3_PIN, OUTPUT);
pinMode(RELAY4_PIN, OUTPUT);

// Default Relays to OFF state (Active LOW Relays)
digitalWrite(RELAY1_PIN, HIGH);
digitalWrite(RELAY2_PIN, HIGH);
digitalWrite(RELAY3_PIN, HIGH);
digitalWrite(RELAY4_PIN, HIGH);

// Initialize PIR Input Sensor
pinMode(PIRPIN, INPUT);

dht.begin();
setup_wifi();
client.setServer(mqtt_server, 1883);
client.setCallback(callback);
}

// =====
// Main Execution Loop
// =====
void loop() {
    if (!client.connected()) {
        reconnect();
    }
    client.loop();

    unsigned long now = millis();
    // Publish telemetry readings every 2000 ms
    if (now - lastMsg > 2000) {
        lastMsg = now;

        float t = dht.readTemperature();
        float h = dht.readHumidity();
        int ldrVal = analogRead(LDRPIN);
        int pirState = digitalRead(PIRPIN);

        if (!isnan(t) && !isnan(h)) {
            char tempString[8];
            dtostrf(t, 1, 2, tempString);
            client.publish("home/temperature", tempString);

            char humString[8];
            dtostrf(h, 1, 2, humString);
            client.publish("home/humidity", humString);
        }

        char ldrString[8];
        itoa(ldrVal, ldrString, 10);
        client.publish("home/light", ldrString);

        client.publish("home/motion", pirState == HIGH ? "DETECTED" : "CLEAR");
    }
}

```